

**A NOVEL RP-HPLC METHOD  
DEVELOPMENT AND VALIDATION  
FOR THE QUANTIFICATION OF  
FUSIDIC ACID**

**SYNOPSIS**

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**Master of Pharmacy  
(Pharmaceutical Chemistry)**

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# 1 Introduction

Fusidic Acid is a steroidal antibiotic extensively used in pharmaceutical formulations for the treatment of bacterial infections caused mainly by Gram-positive organisms, especially *Staphylococcus aureus* infections. It is commonly formulated as topical creams, ointments, eye drops, and oral dosage forms due to its excellent antibacterial activity and therapeutic effectiveness against resistant bacterial strains [1, 2].

The increasing use of antibiotics in pharmaceutical therapy has emphasized the need for maintaining strict quality control standards during manufacturing and storage. The therapeutic efficacy and safety of pharmaceutical products depend largely on their chemical stability, purity, and absence of harmful degradation products. During manufacturing, packaging, transportation, and storage, pharmaceutical compounds are exposed to several environmental stress factors such as heat, light, moisture, oxidation, acidic conditions, and alkaline conditions, which may result in degradation of the active pharmaceutical ingredient (API) [3, 4].

Drug degradation can lead to reduced potency, altered pharmacological activity, and formation of toxic impurities. Therefore, it becomes essential to develop reliable analytical methods capable of separating the active drug from its degradation products and impurities. Stability studies play an important role in determining the shelf life, storage conditions, and overall quality of pharmaceutical formulations [1].

High Performance Liquid Chromatography (HPLC) is one of the most widely accepted and reliable analytical techniques used in pharmaceutical industries for qualitative and quantitative estimation of drugs and impurities. HPLC offers several advantages such as high precision, excellent sensitivity, reproducibility, selectivity, rapid analysis, and efficient separation capability [5, 6].

Reverse phase HPLC methods are commonly employed for the analysis of pharmaceutical compounds because of their simplicity and effectiveness. In recent years, stability-indicating HPLC methods have gained significant importance in pharmaceutical quality control and regulatory submissions. A stability-indicating method is defined as an analytical method that accurately and specifically measures the active pharmaceutical ingredient without interference from degradation products, impurities, or excipients [3].

The present research work focuses on the development and validation of a simple, accurate, precise, and stability-indicating HPLC method for the quantification of fusidic acid and its degradation products.

## **2 Aim and Objectives of the Study**

The main aim of the present research work is to develop and validate a simple, precise, accurate, robust, and stability-indicating reverse phase HPLC method for the quantitative estimation of fusidic acid and its degradation products in pharmaceutical formulations.

The specific objectives of the study are as follows:

1. To perform a comprehensive literature review regarding analytical methods reported for fusidic acid estimation and degradation studies [7].
2. To study the physicochemical properties of fusidic acid relevant to chromatographic analysis [8].
3. To develop suitable chromatographic conditions for effective separation of fusidic acid and its degradation impurities.
4. To optimize chromatographic parameters such as:
  - Mobile phase composition
  - Flow rate
  - Detection wavelength
  - pH of mobile phase
  - Column temperature
5. To carry out forced degradation studies under various stress conditions.
6. To validate the developed analytical method according to ICH guidelines.

## **3 Literature Review**

A detailed literature survey was conducted using various scientific journals, papers,

pharmacopoeias, textbooks, and online research databases related to fusidic acid analysis, pharmaceutical stability studies, and HPLC method development techniques [5]. Several analytical techniques have been reported for the determination of fusidic acid in bulk drugs and pharmaceutical formulations, including spectrophotometry, TLC, microbiological assays, and HPLC. Different research studies reported the use of reverse phase HPLC methods for the estimation of fusidic acid. However, many reported methods possess limitations such as longer retention time, poor peak symmetry, insufficient separation of degradation products, and lack of comprehensive validation [7]. Literature review studies indicate that forced degradation and stability-indicating studies play a major role in pharmaceutical drug development and analytical method validation. In the review [2] the authors explained that forced degradation involves exposing drug substances and products to stress conditions such as acidic, alkaline, oxidative, thermal, humidity, and photolytic environments to generate degradation products within a short period. These studies help identify degradation pathways, determine intrinsic stability, understand degradation mechanisms, and develop stability-indicating analytical methods and suitable packaging systems. The review also suggested that approximately 5–20% degradation is generally acceptable during analytical validation and highlighted the importance of conducting stress studies early in the drug development process. Additionally, the paper discussed the application of analytical techniques such as HPLC, LC-MS, and spectroscopic methods for characterization of degradation products. Another review paper [7] emphasized the significance of Reverse Phase High Performance Liquid Chromatography (RP-HPLC) in pharmaceutical analysis because of its high sensitivity, accuracy, reproducibility, and rapid separation capability. The authors described chromatographic principles, optimization of mobile and stationary phases, sample preparation, and important chromatographic parameters such as retention time, resolution, theoretical plates, and asymmetry factor. The paper also discussed analytical method validation according to ICH guidelines, including parameters like accuracy, precision, specificity, linearity, robustness, limit of detection, and limit of quantification, concluding that RP-HPLC is one of the most reliable techniques for pharmaceutical quality control and stability studies. Furthermore, the ICH guideline [1] provided international regulatory recommendations for stability testing of pharmaceutical products under environmental conditions such as temperature, humidity, and light. The guideline highlighted the importance of long-term, accelerated, and intermediate stability studies to establish shelf life and storage conditions. It also recommended stress testing involving oxidation, hydrolysis, humidity, temperature, and photolysis studies to identify degradation products and degradation pathways. Moreover, the guideline emphasized the necessity of validated stability-indicating analytical

methods, suitable packaging systems, and regular stability monitoring to ensure the safety, quality, and efficacy of pharmaceutical products throughout their shelf life and for regulatory approval purposes.

| Reference paper                                                                                                                                                          | Column Used                                             | Mobile Phase                                                      | Flow Rate             | Retention Time (RT)                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------|-----------------------|----------------------------------------------|
| Development and Validation of a Novel Stability-Indicating RP-HPLC Method for Simultaneous Determination of Halometasone, Fusidic Acid, Methylparaben and Propylparaben” | C18 Column                                              | Gradient RP-HPLC system                                           | 1.0 mL/min            | Fusidic acid RT not specifically mentioned   |
| Development and Validation of HPLC Method for Detection of Fusidic Acid Loaded in Non-ionic and Cationic Nanoemulsion-based Gels                                         | RP-C18 (250 × 4.6 mm, 5 µm)                             | Water : Acetonitrile (1:4), pH 3.3                                | 1.0 mL/min            | Not specifically reported                    |
| RP-HPLC Method Development and Validation for Simultaneous Estimation of Clobetasol Propionate and Fusidic Acid in Cream                                                 | Shim-pack Solar C18 (250 × 4.6 mm, 5 µm)                | Acetonitrile : Water (90:10), pH 5.0                              | 1.0 mL/min            | Fusidic acid = 6.006 min                     |
| Development of a Validated Analytical Method for Estimation of Sodium Fusidate in Pharmaceutical Dosage Form by RP-HPLC                                                  | Thermo Scientific Hypersil ODS C18 (150 × 4.6 mm, 5 µm) | Methanol : 1% Glacial Acetic Acid : Acetonitrile (10:30:60 v/v/v) | 1.1 mL/min            | 5.597 min                                    |
| An RP-HPLC Method for Stability-Indicating Analysis of Impurities of Fusidic Acid and Betamethasone-17-Valerate                                                          | RP-HPLC Column                                          | Stability-indicating mobile phase system                          | Not clearly specified | Not clearly specified                        |
| Assessment of Fusidic Acid and Beclomethasone Dipropionate in Semi-solid Dosage Form Using Validated HPLC Method                                                         | Kromasil C18 (125 × 4.6 mm, 5 µm)                       | Methanol : Orthophosphoric Acid : Acetonitrile (10:45:45 v/v/v)   | 2.5 mL/min            | RT values reported during validating studies |
| Novel Stability Indicating RP-HPLC Method Development                                                                                                                    | Altima C18 (150 × 4.6 mm, 5 µm)                         | Acetonitrile : 0.1% OPA Buffer (50:50 v/v)                        | 1.0 mL/min            | 2.7 min                                      |
| Validated RP-HPLC Method for Fusidic Acid, Hydrocortisone                                                                                                                | Kromasil C18 (150 × 4.6 mm, 5 µm)                       | OPA Buffer and Acetonitrile Gradient                              | 2.0 mL/min            | Not specifically mentioned                   |

|                               |     |        |  |  |
|-------------------------------|-----|--------|--|--|
| Acetate and Potassium Sorbate | μm) | System |  |  |
|-------------------------------|-----|--------|--|--|

## 4 Materials and Instruments Used

### 4.1 Chemicals and Reagents

- Fusidic acid reference standard
- Methanol (HPLC grade)
- Acetonitrile (HPLC grade)
- Water (HPLC grade)
- Orthophosphoric acid
- Hydrochloric acid
- Sodium hydroxide
- Hydrogen peroxide

### 4.2 Instruments Used

- HPLC system equipped with UV detector
- Analytical balance
- Digital pH meter
- Ultrasonic bath
- Vacuum filtration assembly

- UV chamber

- Hot air oven

### **4.3 Chromatographic Column**

A reverse phase C18 column was selected for chromatographic separation because of its high efficiency and reproducibility [6].

## **5 Work Completed So Far**

### **5.1 Literature Survey**

An extensive literature review was carried out using journals, research articles, pharmacopoeias, and online databases.

### **5.2 Procurement of Chemicals and Reagents**

Required chemicals, reagents, solvents, and standards were procured.

### **5.3 Instrumental Setup and Calibration**

The HPLC system and associated laboratory instruments were calibrated and verified.

## **6 Work in Progress**

### **6.1 Preliminary Method Development Trials**

Several chromatographic trials were conducted using different mobile phase combinations and flow rates.

### **6.2 Optimization of Chromatographic Conditions**

Optimization studies were completed successfully.

### **6.3 Forced Degradation Studies**

Forced degradation studies under various stress conditions were completed successfully.

### **6.4 Method Validation Studies**

Validation studies were performed according to ICH guidelines.

### **6.5 Statistical Analysis of Validation Data**

Detailed statistical evaluation of validation parameters is currently under progress.

### **6.6 Compilation of Chromatograms and Tables**

Chromatographic figures, degradation chromatograms, calibration curves, and validation tables are being compiled.

### **6.7 Interpretation of Degradation Behavior**

Detailed interpretation of degradation pathways is under progress.

### **6.8 Thesis Writing and Documentation**

Preparation of final report chapters including results, discussion, references, figures, and tables is in progress.

## **7 Future Scope**

### **7.1 Routine Pharmaceutical Quality Control**

The validated HPLC method can be routinely used for quantitative estimation of fusidic acid in pharmaceutical dosage forms.

### **7.2 Stability Testing Studies**

The stability-indicating capability of the method makes it useful for stability testing studies.



### **7.3 Detection of Degradation Impurities**

The method can be applied for identification and monitoring of degradation impurities.

### **7.4 Regulatory Analytical Applications**

The validated analytical procedure can be used for regulatory submissions and pharmaceutical documentation.

### **7.5 Advanced Analytical Research**

Further studies may involve advanced analytical techniques such as LC-MS and NMR spectroscopy.

## **8 Conclusion**

The present research work entitled “Quantification and Validation of Fusidic Acid and Its Degradation Products by HPLC” has progressed successfully according to the proposed research objectives.

A simple, accurate, precise, robust, and stability-indicating reverse phase HPLC method has been successfully developed and optimized.

Forced degradation studies confirmed the specificity and stability-indicating capability of the method.

Validation studies confirmed that the developed method is accurate, precise, specific, sensitive, robust, and reproducible.

The method can be effectively applied for routine pharmaceutical quality control analysis, stability studies, impurity profiling, and regulatory analytical applications.

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